Package 'manMetaVAR'

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From Person-Specific Dynamics to Population Inference:
A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework
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<doi:10.0000/0000000000>.

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Author Ivan Jacob Agaloos Pesigan [aut, cre, cph] (ORCID:
<`https://orcid.org/0000-0003-4818-8420`>),
Michael A. Russell [ctb] (ORCID:
<`https://orcid.org/0000-0002-3956-604X`>),
Sy-Miin Chow [ctb] (ORCID: <`https://orcid.org/0000-0003-1938-027X`>)

Maintainer Ivan Jacob Agaloos Pesigan <`r.jeksterslab@gmail.com`>

Contents

`Check` *Check Replication*

Description

Check Replication

Usage

```
Check(taskid, repid, output_folder, naive, metavar, mplus)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`naive`	Logical. If `naive = TRUE`, fit the naive approach.
`metavar`	Logical. If `metavar = TRUE`, fit the model using the `metaDyn` package.
`mplus`	Logical. If `mplus = TRUE`, fit the model using DSEM in `Mplus`.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`CheckFitDTVAR` *Check Replication - FitDTVAR*

Description

Check Replication - FitDTVAR

Usage

```
CheckFitDTVAR(taskid, repid, output_folder, suffix)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.

Details

This function is executed via the `Check` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`CheckFitMetaVAR` *Check Replication - FitMetaVAR*

Description

Check Replication - FitMetaVAR

Usage

```
CheckFitMetaVAR(taskid, repid, output_folder, suffix)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.

Details

This function is executed via the `Check` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`CheckFitMplus` *Check Replication - FitMplus*

Description

Check Replication - FitMplus

Usage

```
CheckFitMplus(taskid, repid, output_folder, suffix)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.

Details

This function is executed via the `Check` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`CheckFitNaive` *Check Replication - FitNaive*

Description

Check Replication - FitNaive

Usage

```
CheckFitNaive(taskid, repid, output_folder, suffix)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.

Details

This function is executed via the `Check` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`Compress` *Compress Replication*

Description

Compress Replication

Usage

```
Compress(taskid, repid, output_folder)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

FigMetricHeatmap *Results Heatmap*

Description

Plot results heatmap.

Usage

```
FigMetricHeatmap(results, metric_name, grey_scale = FALSE, values = FALSE)
```

Arguments

`results`	`results` data frame.
`metric_name`	Character string. `"coverage"` for coverage probability, `"abs_rel_bias"` for absolute value of the relative bias, `"rmse"` for RMSE, and `"power"` for statistical power.
`grey_scale`	Logical. If `TRUE`, use a grey-scale fill suitable for black-and-white printing.
`values`	Logical. If `TRUE`, display values inside tiles.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: `FigOverview()`

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigMetricHeatmap(
  results = results,
  metric_name = "coverage"
)

## End(Not run)
```

FigOverview *Results Overview*

Description

Plot results overview.

Usage

```
FigOverview(results)
```

Arguments

results	results data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: FigMetricHeatmap()

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigOverview(results)

## End(Not run)
```

FitDTVAR *Fit the Model using the fitVARMxID Package*

Description

The function fits the model using the fitVARMxID package.

Usage

```
FitDTVAR(data, ncores = NULL, seed = NULL)
```

Arguments

data	R object. Output of the GenData() function.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: FitMetaVAR(), FitMplus(), FitNaive(), MplusInput()

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

FitMetaVAR *Multivariate Meta-Analysis using the metaDyn Package*

Description

The function performs multivariate meta-snalysis using the metaDyn package.

Usage

```
FitMetaVAR(fit, ncores = NULL, seed = NULL)
```

Arguments

fit	R object. Output of the FitDTVAR() function.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: FitDTVAR(), FitMplus(), FitNaive(), MplusInput()

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
meta <- FitMetaVAR(fit = fit, seed = seed)
summary(meta)
print(meta)
coef(meta)
```

```
vcov(meta)

## End(Not run)
```

FitMplus *Fit the Model using Mplus*

Description

The function fits the model using Mplus.

Usage

```
FitMplus(
  data,
  chains = 2L,
  iter = 40000L,
  fscores = NULL,
  plot = FALSE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL,
  seed = NULL
)
```

Arguments

data	R object. Output of the `GenData()` function.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.
wd	Character string. Working directory.
mplus_bin	Character string. Path to Mplus binary. If `mplus_bin = NULL`, the function will try to find the appropriate binary.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMetaVAR()`, `FitNaive()`, `MplusInput()`

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitMplus(data = data)
print(fit)
summary(fit)

## End(Not run)
```

`FitNaive` *Naive*

Description

The function performs the naive "fit-many-then-summarize".

Usage

```
FitNaive(fit, seed = NULL)
```

Arguments

`fit`	R object. Output of the `FitDTVAR()` function.
`seed`	Integer. Random seed.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMetaVAR()`, `FitMplus()`, `MplusInput()`

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
naive <- FitNaive(fit = fit, seed = seed)
summary(naive)
print(naive)
coef(naive)
vcov(naive)

## End(Not run)
```

GenData *Simulate Data*

Description

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid, seed = NULL)
```

Arguments

`taskid`	Positive integer. Task ID.
`seed`	Integer. Random seed.

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
print(data)
summary(data)
plot(data)

## End(Not run)
```

manmetavar-data-methods *Methods for Objects of Class* `manmetavar.data`

Description

This page documents the available methods for objects of class `manmetavar.data`.

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.data'
summary(object, ...)

## S3 method for class 'manmetavar.data'
plot(x, ...)
```

Arguments

`x`	Object of class `manmetavar.data`.
`...`	additional arguments.
`object`	Object of class `manmetavar.data`.

Author(s)

Ivan Jacob Agaloos Pesigan

`manmetavar-dtvar-methods`

Methods for Objects of Class `manmetavar.dtvar`

Description

This page documents the available methods for objects of class `manmetavar.dtvar`.

Usage

```
## S3 method for class 'manmetavar.dtvar'
coef(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
vcov(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  robust = FALSE,
  ncores = NULL,
  ...
)
```

```
## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
  ...
)
```

Arguments

`object`	Object of class `manmetavar.dtvar`.
`mu`	Logical. If `mu = TRUE`, include estimates of the `mu` vector, if available. If `mu = FALSE`, exclude estimates of the `mu` vector.
`alpha`	Logical. If `alpha = TRUE`, include estimates of the `alpha` vector, if available. If `alpha = FALSE`, exclude estimates of the `alpha` vector.
`beta`	Logical. If `beta = TRUE`, include estimates of the `beta` matrix, if available. If `beta = FALSE`, exclude estimates of the `beta` matrix.
`nu`	Logical. If `nu = TRUE`, include estimates of the `nu` vector, if available. If `nu = FALSE`, exclude estimates of the `nu` vector.
`psi`	Logical. If `psi = TRUE`, include estimates of the `psi` matrix, if available. If `psi = FALSE`, exclude estimates of the `psi` matrix.
`theta`	Logical. If `theta = TRUE`, include estimates of the `theta` matrix, if available. If `theta = FALSE`, exclude estimates of the `theta` matrix.
`ncores`	Positive integer. Number of cores to use.

`...`	additional arguments.
`robust`	Logical. If `TRUE`, use robust (sandwich) sampling variance-covariance matrix. If `FALSE`, use normal theory sampling variance-covariance matrix.
`x`	Object of class `manmetavar.dtvar`.
`means`	Logical. If `means = TRUE`, return means. Otherwise, the function returns raw estimates.
`digits`	Integer indicating the number of decimal places to display.

Author(s)

Ivan Jacob Agaloos Pesigan

`manmetavar-metavar-methods`

Methods for Objects of Class `manmetavar.metavar`

Description

This page documents the available methods for objects of class `manmetavar.metavar`.

This page documents the available methods for objects of class `manmetavar.naive`.

Usage

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.metavar'
vcov(object, robust = NULL, ...)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
summary(object, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, robust = NULL, ...)

## S3 method for class 'manmetavar.naive'
coef(object, ...)

## S3 method for class 'manmetavar.naive'
vcov(object, ...)

## S3 method for class 'manmetavar.naive'
```

```
print(x, ...)

## S3 method for class 'manmetavar.naive'
summary(object, alpha = 0.05, digits = 4, ...)
```

Arguments

`object`	Object of class `manmetavar.naive`.
`...`	additional arguments.
`robust`	Logical. If `TRUE`, use robust (sandwich) sampling variance-covariance matrix. If `FALSE`, use normal theory sampling variance-covariance matrix. If `NULL`, the function will check `object` if robust standard errors are available.
`x`	Object of class `manmetavar.naive`.
`alpha`	Numeric vector. Significance level α.
`digits`	Integer indicating the number of decimal places to display.
`parm`	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
`level`	the confidence level required.

Author(s)

Ivan Jacob Agaloos Pesigan

`manmetavar-mplus-methods`

Methods for Objects of Class `manmetavar.mplus`

Description

This page documents the available methods for objects of class `manmetavar.mplus`.

Usage

```
## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)
```

```
## S3 method for class 'manmetavar.mplus'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,
  burnin = NULL,
  legend_loc = "topright",
  ...
)
```

Arguments

`object`	Object of class `manmetavar.mplus`.
`median`	Logical. If `median = TRUE`, return median of the posterior. If `median = FALSE`, return mean of the posterior.
`burnin`	Integer indicating initial samples to discard. If `burnin = NULL`, do not discard anything.
`...`	additional arguments.
`alpha`	Numeric vector. Significance level α.
`digits`	Integer indicating the number of decimal places to display.
`parm`	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
`level`	the confidence level required.
`x`	Object of class `manmetavar.mplus`.
`what`	Character string. If `what = "posterior"`, return posterior distribution plots. If `what = "trace"`, return trace plots.
`legend_loc`	Character string. Legend location.

Author(s)

Ivan Jacob Agaloos Pesigan

`model` *Model Parameters*

Description

Model Parameters

Usage

```
data(model)
```

Format

A list with 15 elements:

k Number of variables.

mu_mu Mean of the set-point vector $\boldsymbol{\mu}$.

mu_sigma Covariance matrix of the parameter $\boldsymbol{\mu}$.

mu_sigma_l Cholesky factor of the covariance matrix of the parameter $\boldsymbol{\mu}$.

beta_mu Mean of lagged coefficients matrix $\boldsymbol{\beta}$.

beta_sigma Covariance matrix of the parameter $\mathrm{vec}\left(\boldsymbol{\beta}\right)$.

beta_sigma_l Cholesky factor of the covariance matrix of the parameter $\mathrm{vec}\left(\boldsymbol{\beta}\right)$.

psi Process noise covariance matrix $\boldsymbol{\Psi}$.

psi_l Cholesky factor of the process noise covariance matrix $\boldsymbol{\Psi}$.

psi_d_ldl `uc_d` of the LDL' decomposition of the process noise covariance matrix $\boldsymbol{\Psi}$. See `fitVARMxID::LDL()`.

psi_l_ldl `s_l` of the LDL' decomposition of the process noise covariance matrix $\boldsymbol{\Psi}$. See `fitVARMxID::LDL()`.

ma_fixed Vector of fixed effects $\boldsymbol{\theta} = \left[\boldsymbol{\mu}, \mathrm{vec}\left(\boldsymbol{\beta}\right)\right]'$

ma_random Matrix of random effects $\boldsymbol{\theta} = \left[\boldsymbol{\mu}, \mathrm{vec}\left(\boldsymbol{\beta}\right)\right]'$

ma_random_d_ldl `uc_d` of the LDL' decomposition of the random effects. See `fitVARMxID::LDL()`.

ma_random_l_ldl `s_l` of the LDL' decomposition of the random effects. See `fitVARMxID::LDL()`.

Author(s)

Ivan Jacob Agaloos Pesigan

`MplusInput` *Create Mplus Input File*

Description

The function creates an Mplus input file.

Usage

```
MplusInput(
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,
  fscores,
  plot,
  ncores = NULL,
  seed = NULL
)
```

Arguments

- `fn_data` — Character string. Filename for data file.
- `fn_posterior` — Character string. Filename for posterior output.
- `fn_factorscores` — Character string. Filename for factor scores output.
- `chains` — Integer. Number of chains.
- `iter` — Integer. Number of iterations.
- `fscores` — Integer. Number of iterations for factor scores.
- `plot` — Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.
- `ncores` — Positive integer. Number of cores to use.
- `seed` — Integer. Random seed.

See Also

Other Model Fitting Functions: `FitDTVAR()`, `FitMetaVAR()`, `FitMplus()`, `FitNaive()`

Examples

```
cat(
  MplusInput(
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
    chains = 2L,
    iter = 40000L,
    fscores = 1000L,
    plot = TRUE,
    ncores = NULL,
    seed = 42
  )
)
```

---

params — *Simulation Parameters*

Description

Simulation Parameters

Usage

```
data(params)
```

Format

A dataframe with 12 rows and 3 columns:

taskid Simulation Task ID.

n Sample size.

time Number of measurement occassions. If `NA`, measurement occasions vary across IDs.

Author(s)

Ivan Jacob Agaloos Pesigan

`results` *Simulation Results*

Description

Simulation Results

Usage

```
data(results)
```

Format

A dataframe with 912 rows and 24 columns:

taskid Simulation Task ID.

replications Number of replications.

parnames Parameter names.

parameter Population parameter value.

method Method used to fit the data.

n Sample size.

time Number of measurement occassions.

ci Confidence/credible intervals method.

est Mean parameter estimate.

se Mean standard error.

t Mean test statistic.

p Mean p-value.

ll Mean lower limit of the 95% confidence/credible interval.

ul Mean upper limit of the 95% confidence/credible interval.

sig Proportion of statistically significant results.

zero_hit Proportion of replications where the confidence intervals contained zero.

theta_hit Proportion of replications where the confidence intervals contained the population parameter.

sq_error Mean squared error.

bias Bias.

rel_bias Relative bias.

rmse Root mean square error.

coverage Coverage probability.

power Statistical power.

par_idx Parameter index.

Author(s)

Ivan Jacob Agaloos Pesigan

`Sim` *Simulation Replication*

Description

Simulation Replication

Usage

```
Sim(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  seed,
  data,
  naive,
  metavar,
  mplus,
  chains,
  iter,
  fscores,
  plot
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`seed`	Integer. Random seed.
`data`	Logical. Simulate data.
`naive`	Logical. If `naive = TRUE`, fit the naive approach.
`metavar`	Logical. If `metavar = TRUE`, fit the model using the `metaDyn` package.
`mplus`	Logical. If `mplus = TRUE`, fit the model using DSEM in `Mplus`.
`chains`	Integer. Number of chains.
`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitDTVAR` *Simulation Replication - FitDTVAR*

Description

Simulation Replication - FitDTVAR

Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitMetaVAR` *Simulation Replication - FitMetaVAR*

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitMplus` *Simulation Replication - FitMplus*

Description

Simulation Replication - FitMplus

Usage

```
SimFitMplus(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`chains`	Integer. Number of chains.

`iter`	Integer. Number of iterations.
`fscores`	Integer. Number of iterations for factor scores.
`plot`	Logical. If `plot = TRUE`, add `PLOT: TYPE = PLOT3;` to `Mplus` input file.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitNaive` *Simulation Replication - FitNaive*

Description

Simulation Replication - FitNaive

Usage

```
SimFitNaive(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFN` *Simulation File Name*

Description

Simulation File Name

Usage

```
SimFN(output_type, output_folder, suffix)
```

Arguments

`output_type`	Character string. Output type. Valid values include `"data"`, `"fit-dt-var-mx"`, `"fit-meta-var-mx"`, and `"fit-mplus"`.
`output_folder`	Character string. Output folder.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.

Value

Returns a character string file name with the `output_folder` in the OS-specific format.

`SimGenData` *Simulation Replication - GenData*

Description

Simulation Replication - GenData

Usage

```
SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimProj` *Simulation Project Name*

Description

Simulation Project Name

Usage

```
SimProj()
```

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

`Sum` *Summary*

Description

Summary

Usage

```
Sum(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  naive,
  metavar_normal,
  metavar_robust,
  mplus,
  ncores
)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`naive`	Logical. If `naive = TRUE`, summarize naive estimates.
`metavar_normal`	Logical. If `metavar_normal = TRUE`, summarize metaVAR model using normal confidence intervals.
`metavar_robust`	Logical. If `metavar_robust = TRUE`, summarize metaVAR model using robust (sandwich) confidence intervals.
`mplus`	Logical. If `mplus = TRUE`, summarize the DSEM model.
`ncores`	Positive integer. Number of cores to use.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARNormal *Summary: Normal Theory (FitMetaVAR)*

Description

Summary: Normal Theory (FitMetaVAR)

Usage

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARRobust *Summary: Robust (FitMetaVAR)*

Description

Summary: Robust (FitMetaVAR)

Usage

```
SumFitMetaVARRobust(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`ncores`	Positive integer. Number of cores to use.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitMplus` *Summary (FitMplus)*

Description

Summary (FitMplus)

Usage

```
SumFitMplus(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`ncores`	Positive integer. Number of cores to use.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SumFitNaive` *Summary: Naive*

Description

Summary: Naive

Usage

```
SumFitNaive(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`ncores`	Positive integer. Number of cores to use.

Details

This function is executed via the `Sum` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

Index